GM UNIVERSITY

(Established under the Karnataka State Act No. 19 of 2023)
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Faculty of Engineering and Technology

Department of Computer Science and Engineering

Subject: Artificial Intelligence and Machine Learning

Subject Code: UE24CS2401

Question Bank

Module 1

Syllabus:

Introduction to Artificial intelligence:

What is AI: How Does AI Work, Advantages and Disadvantages of Artificial Intelligence

History of AI

Types of AI: Weak AI or Narrow AI, General AI, Strong AI/ Super AI, Reactive Machines, Limited Memory , Theory of Mind, Self-Awareness

Components of Intelligence: Agent and Environment,Types of agents

Informed Search Algorithms: Pure Heuristic, Best first, A*, AO* Algorithms.

CO2 Level Questions

1. Explain the concept of Artificial Intelligence with suitable examples.
2. Describe how Artificial Intelligence works in real-world applications.
3. Summarize the advantages and Disadvantages of Artificial Intelligence in modern society.
4. Summarize the major milestones in the evolution(history) of Artificial Intelligence.
5. Differentiate between Narrow AI and General AI.

CO3 Level Questions

1. Apply the Best First Search algorithm to solve a given graph problem and trace the steps.
2. Solve a pathfinding problem using the A* algorithm and explain how f(n) is calculated.
3. Apply the AO* algorithm to solve an AND-OR graph and identify the optimal solution path

CO4 Level Questions

1. Analyze the evolution of AI from its early stages to modern deep learning systems. What key technological changes influenced its growth?
2. Analyze how Weak AI differs from strong AI systems and what challenges exist in implementing it.
3. Analyze the different types of Agents in AI and its environment. Identify percepts, actions and goals with an example.

Module 3:

Module 3:

Introduction to Machine Learning : Need for Machine Learning, Machine Learning Explained

Machine Learning in Relation to other Fields: Machine Learning and Artificial Intelligence, Machine Learning, Data Science, Data Mining, and Data Analytics , Machine Learning and Statistics

Types of Machine Learning: Supervised Learning: Classification and regression, Unsupervised Learning, Semi-supervised Learning, Reinforcement learning, MACHINE LEARNING PROCESS, challenges, Applications

Understanding Data: what is Data (TYPES OF DATA), BIG DATA ANALYTICS AND TYPES OF ANALYTICS , BIG DATA ANALYSIS FRAMEWORK:DATA COLLECTION , DATA PREPROCESSING, Univariate Data Analysis and Visualization(Data Visualization, Central Tendency, Dispersion). Bivariate Data (Bivariate Statistics), Feature Engineering and Dimensionality Reduction Techniques (Step wise Forward and Backward).

CO2 Level Questions

1. Explain the need for Machine Learning in solving real-world problems.
2. Explain the relationship between Machine Learning and Artificial Intelligence.
3. Explain Supervised Learning with a suitable example
4. Describe the steps involved in the Machine Learning process.
5. Explain the differences between structured, semi-structured, and unstructured data with examples.
6. Explain the difference between descriptive, predictive, and prescriptive analytics.
7. Differentiate between univariate and bivariate data

CO3 Level questions

1. Apply the concept of data visualization to show how sales trends can be represented using Bar chart, Pie chart, Histogram, Dot Plots.
2. Apply the concept of dispersion for the patients list {12, 14, 19, 22, 24, 28, 31, and 34} and find IQR & Five point summary.

CO4 Level Questions

1. Analyze the role of Normalization procedure: Min-Max and Z-score in data preprocessing for given set V= {88, 90, 92, 94} for range 0-1 and V= {10, 20, 30} respectively.

2. Calculate Covariance and Correlation for the data X={1,2,3,4,5} and Y={1,8,27,64,125} using concept of Bivariate statistics.